

ICPC Team Notebook

typedef unsigned long long ll;
Sorbonne Université
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A curated reference of algorithms and data structures



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1 Data Structures

1.1 Fenwick Tree

Point update, prefix and range sum, $O(\log n)$, $O(n)$ build

```
struct Fenwick {
    int n; vector<ll> t; //using long long, int might overflow.
    Fenwick(vector<ll>& a): n(a.size()), t(n+1,0) {
        for(int i = 1; i <= n; i++) { // Builds tree from array in O(n)
            t[i] += a[i-1];
            int p = i+(i&-i);
            if(p<=n) t[p] += t[i];
        }
    }
    void add(int i,long long v){
        for(;i<=n;i+=i&-i) t[i]+=v;
    }
    long long sum(int i){
        long long r=0;
        for(; i>0 ;i -= i&-i) r += t[i];
        return r;
    }
    long long sum(int l,int r){ return sum(r)-sum(l-1); }
};
```

1.2 FenwickPython

```
class Fenwick:
    def __init__(self, t):
        self.s = [0] * (len(t) + 1) # create internal storage
        for a, v in enumerate(t):
            self.add(a, v) # initialize
    def prefixSum(self, a):
        i = a + 1 # internal index starts at 1
        total = 0
        while i > 0: # loops over neighbors
            total += self.s[i] # cumulative sum
            i -= (i & -i) # left neighbor
        return total
    def intervalSum(self, a, b):
        return self.prefixSum(b) - self.prefixSum(a-1)
    def add(self, a, val):
        i = a + 1 # internal index starts at 1
        while i < len(self.s): # loops over parents
            self.s[i] += val # update node
            i += (i & -i) # parent
```

1.3 RMQ

Preprocess an array in $O(n \log n)$ to answer range minimum queries in $O(1)$

```
struct RMQ {
    vvi jmp;
    RMQ(const vi& V) { // Python: depth = N.bit_length()
        int N = V.size(), depth = 32 - __builtin_clz(N);
        jmp.assign(depth, V);
        for(int i = 0; i < depth-1; ++i)
            for(int j = 0; j < N; ++j)
                jmp[i+1][j] = min(jmp[i][j], jmp[i][min(N-1, j + (1<<i))]);
    } // Returns min{V[a], V[a+1], ..., V[b-1]}
    int query(int a, int b) {
```

```

    if(b <= a) return INT_MAX;
    int dep = 31 - __builtin_clz(b - a); // python: (b-a).bit_length() - 1
    return min(jmp[dep][a], jmp[dep][b - (1<<dep)]);
}
};

```

1.4 Segment Tree(s)

Regular seg tree: point update, range sum query, $O(\log n)$

```

struct SegTree {
    int n; vector<ll> t;
    SegTree(vi &a): n(a.size()), t(4*n) { build(a,1,0,n); }
    void build(vi &a,int v,int l,int r){
        if(r-l == 1){ t[v]=a[l]; return; }
        int m = (l+r)/2;
        build(a, 2*v, l, m); build(a, 2*v+1, m, r);
        t[v] = t[2*v] + t[2*v+1];
    }
    ll sum_aux(int v,int l,int r,int ql,int qr){
        if(qr <= l || r <= ql) return 0;
        if(ql <= l && r <= qr) return t[v];
        int m = (l+r)/2;
        return sum_aux(2*v, l,m,ql,qr) + sum_aux(2*v+1, m,r,ql,qr);
    }
    ll sum(int ql,int qr){ return sum_aux(1,0,n,ql,qr); }
    void upd_aux(int v,int l,int r,int pos,ll val){
        if(r-l==1){ t[v]=val; return; }
        int m = (l+r)/2;
        if(pos<m) upd_aux(2*v,l,m,pos,val);
        else upd_aux(2*v+1,m,r,pos,val);
        t[v]=t[2*v]+t[2*v+1];
    }
    void upd(int pos,ll val){ upd_aux(1,0,n,pos,val); }
};

```

Lazy SegTree: range add, range sum query, $O(\log n)$

```

struct LazySegTree {
    int n; vector<ll> t, lazy;
    LazySegTree(vi &a):n(a.size()),t(4*n),lazy(4*n){build(a,1,0,n);}
    void build(vi &a,int v,int l,int r){
        if(r-l == 1){ t[v]=a[l]; return; }
        int m = (l+r)/2;
        build(a,2*v,l,m); build(a,2*v+1,m,r);
        t[v] = t[2*v]+t[2*v+1];
    }
    void push(int v,int l,int r){
        if(lazy[v]==0) return;
        t[v] += lazy[v]*(r-l);
        if(r-l>1){
            lazy[2*v] += lazy[v];
            lazy[2*v+1] += lazy[v];
        }
        lazy[v]=0;
    }
    void upd_aux(int v,int l,int r,int ql,int qr,ll val){
        push(v,l,r);
        if(qr<=l || r<=ql) return;
        if(ql<=l && r<=qr){
            lazy[v] += val; push(v,l,r);
            return;
        }
        int m = (l+r)/2;
        upd_aux(2*v, l,m,ql,qr,val);
        upd_aux(2*v+1, m,r,ql,qr,val);
        t[v] = t[2*v] + t[2*v+1];
    }
};

```

```

}
ll sum_aux(int v,int l,int r,int ql,int qr){
    push(v,l,r);
    if(qr<=l || r<=ql) return 0;
    if(ql<=l && r<=qr) return t[v];
    int m = (l+r)/2;
    return sum_aux(2*v, l,m,ql,qr) + sum_aux(2*v+1, m,r,ql,qr);
}
void upd(int ql,int qr,ll val){ upd_aux(1,0,n,ql,qr,val); }
ll sum(int ql,int qr){ return sum_aux(1,0,n,ql,qr); }
};

```

Iterative SegTree: point update, range sum query, $O(\log n)$

```

struct IterSegTree {
    int n; vector<ll> t;
    IterSegTree(vi &a): n(a.size()), t(2*n) {
        for(int i=0;i<n;i++) t[n+i]=a[i];
        for(int i=n-1;i>0;i--) t[i]=t[i<<1]+t[i<<1|1];
    }
    void upd(int pos,ll val){
        pos+=n; t[pos]=val;
        for(pos>>=1; pos>0; pos>>=1) t[pos]=t[pos<<1]+t[pos<<1|1];
    }
    ll sum(int l,int r){
        ll res=0;
        for(l+=n,r+=n;l<r;l>>=1,r>>=1){
            if(l&1) res+=t[l++];
            if(r&1) res+=t[--r];
        }
        return res;
    }
};

```

1.5 Trie

Prefix tree: stores strings, insert/find in $O(|s|)$; handles lowercase letters a-z

```

struct Node{
    int next[26]; bool isEnd=false;
    Node(){fill(begin(next), end(next), -1);}
};
struct Trie{
    vector<Node> t;
    Trie() {t.emplace_back();}
    void insert(string s){
        int cur = 0;
        for (auto c : s){
            if (t[cur].next[c-'a'] == -1){
                t.emplace_back(); // Adds new node
                t[cur].next[c-'a'] = t.size() - 1;
            }
            cur = t[cur].next[c-'a'];
        }
        t[cur].isEnd = true;
    }
    bool find(string s){
        int cur = 0;
        for (auto c : s){
            if (t[cur].next[c-'a'] == -1) return false;
            cur = t[cur].next[c-'a'];
        }
        return t[cur].isEnd;
    }
};

```

1.6 Union find

Complexity: effectively $O(1)$

```
struct union_find{
    vector<int> rank, parent;
    union_find(int n){
        rank.resize(n, 0); parent.resize(n);
        for (int i = 0; i < n; i++) parent[i] = i;
    }
    int find(int i){
        int root = parent[i];
        if (parent[root] != root) return parent[i] = find(root);
        return root;
    }
    void unite(int x, int y) {
        int xRoot = find(x);
        int yRoot = find(y);
        if (xRoot == yRoot) return;
        if (rank[xRoot] < rank[yRoot]) parent[xRoot] = yRoot;
        else if (rank[yRoot] < rank[xRoot]) parent[yRoot] = xRoot;
        else{
            parent[yRoot] = xRoot;
            rank[xRoot]++;
        }
    }
};
```

2 Graph Algorithms

2.1 2-SAT

Check if n variables satisfy $(x_i \vee x_j) \wedge (\neg x_k \vee x_l) \wedge \dots$

```
struct TwoSAT {
    int n; SCC g; // c.f. strongly connected components
    TwoSAT(int vars): n(vars), g(2*vars) {}
    int var(int i){ return i; }
    int neg(int i){ return i+n; }
    void add_clause(int a, bool a_val, int b, bool b_val){
        int u = a_val ? var(a) : neg(a);
        int v = b_val ? var(b) : neg(b);
        g.add_edge(u^1, v); // ~a -> b. ^ bitwise XOR, same in python
        g.add_edge(v^1, u); // ~b -> a
    }
    bool satisfiable(vi& ans){
        g.kosaraju();
        ans.assign(n,0);
        for(int i=0;i<n;i++){
            if(g.scc[var(i)] == g.scc[neg(i)]) return false;
            ans[i] = g.scc[var(i)] > g.scc[neg(i)];
        }
        return true;
    }
};
```

2.2 Articulation Points and Bridges

Biconnected components: maximal subgraph that stays connected if any one vertex is removed.
 Articulation points connect multiple BCCs; edges in each child subtree of an AP form a BCC.
 $O(V + E)$

AP = vertices whose removal increases # of connected components.

Bridge = edges whose removal increases # of connected components.

```
void tarjan(int u, vvi &adj, vi &vis, vi &disc, vi &low,
            int &time, int p, vi &isAP, set<pii> &bridges) {
    vis[u] = 1;
```

```
    disc[u] = low[u] = ++time;
    int children = 0;
    for (int v : adj[u]) {
        if (v == p) continue;
        if (!vis[v]) {
            children++;
            tarjan(v, adj, vis, disc, low, time, u, isAP, bridges);
            low[u] = min(low[u], low[v]);
            if (low[v] > disc[u]) // Bridge
                bridges.insert({min(u, v), max(u, v)});
            if (p != -1 && low[v] >= disc[u])
                isAP[u] = 1; // Articulation Point
        }
        else low[u] = min(low[u], disc[v]);
    }
    if (p == -1 && children > 1)
        isAP[u] = 1;
}

pair<vi, set<pii>> findAPandBridges(int V, vvi &adj) {
    vi disc(V), low(V), vis(V), isAP(V); // Initialize with 0s
    set<pii> bridges;
    int time = 0;
    for (int i = 0; i < V; i++)
        if (!vis[i])
            tarjan(i, adj, vis, disc, low, time, -1, isAP, bridges);
    return {isAP, bridges};
}
```

2.3 Cycle detection

Returns a cycle in a directed graph, or empty if none exists. $O(n + m)$

```
vi dfs_cycle(const vvi &adj) {
    int n = adj.size();
    vi color(n, 0), path;
    vi cycle;
    function<bool(int)> dfs = [&](int v) {
        color[v] = 1; path.push_back(v);
        for (int u : adj[v]) {
            if (color[u] == 0 && dfs(u)) return true;
            else if (color[u] == 1) {
                auto it = find(path.begin(), path.end(), u);
                cycle.assign(it, path.end());
                return true;
            }
        }
        path.pop_back(); color[v] = 2;
        return false;
    };
    for (int i = 0; i < n; i++)
        if (color[i] == 0 && dfs(i)) break;
    return cycle;
}
```

Check if undirected graph contains a cycle in $O(m)$.

```
bool dsu_cycle(int n, vvi &edges) {
    union_find uf(n); //c.f. union find
    for (auto &e : edges) {
        if (uf.find(e[0]) == uf.find(e[1])) return true;
        uf.unite(e[0], e[1]);
    }
    return false;
}
```

Tortoise & Hare: finds start of cycle in a functional graph (-1 if none). $O(n)$

```
int floyd_cycle(int start, vi& nxt){
```

```

int tort=start, hare=start;
do {
    if(hare==-1 || nxt[hare]==-1) return -1;
    tort = nxt[tort];
    hare = nxt[nxt[hare]];
} while(tort!=hare);
tort=start;
while(tort!=hare){
    tort=nxt[tort]; hare=nxt[hare];
}
return tort;
}

```

2.4 Diameter of a Tree

Computes the length of the longest path between any two nodes in $O(n)$

```

pii dfs(int u, int p, vvi& adj) {
    pair<int,int> res = {0, u}; // {distance, node}
    for(int v : adj[u]) if(v != p) {
        auto t = dfs(v, u, adj);
        t.first++;
        res = max(res, t);
    }
    return res;
}
int diameter(vvi& adj)
{
    return dfs(dfs(0, -1, adj).second, -1, adj).first;
}

```

2.5 DijkstraPython

SSSP $O((V + E) \log V)$

```

def dijkstra(graph, weight, source=0, target=None):
    n = len(graph)
    assert all(weight[u][v] >= 0 for u in range(n) for v in graph[u])
    prec = [None] * n
    black = [False] * n
    dist = [float('inf')] * n
    dist[source] = 0
    heap = [(0, source)]
    while heap:
        dist_node, node = heappop(heap) # Closest node from source
        if not black[node]:
            black[node] = True
        if node == target:
            break
        for neighbor in graph[node]:
            dist_neighbor = dist_node + weight[node][neighbor]
            if dist_neighbor < dist[neighbor]:
                dist[neighbor] = dist_neighbor
                prec[neighbor] = node
                heappush(heap, (dist_neighbor, neighbor))
    return dist, prec

```

2.6 Euler paths

Euler Path/Circuits:

Undirected:

- Circuit: all vertices even degree and connected.

- Path: exactly 2 odd-degree vertices and connected; start at one of them.

- Trick: to use circuit function for path, add dummy edge between odd vertices, find circuit, remove dummy.

Directed:

- Circuit: strongly connected, $\text{indegree}[v] == \text{outdegree}[v] \forall v$.

- Path: one vertex with $\text{out-in}=1$ (start), one with $\text{in-out}=1$ (end), rest balanced.

- Trick: add dummy edge end-;start, find circuit, rotate to remove dummy.

Find Euler path in directed graph. $O(E)$ MODIFIES THE GRAPH!!!

```

vi euler_directed(vvi &adj, int start=0) {
    vi path; stack<int> st; st.push(start);
    while(!st.empty()) {
        int v = st.top();
        if(!adj[v].empty()) {
            int u = adj[v].back(); adj[v].pop_back();
            st.push(u);
        } else {
            path.push_back(v);
            st.pop();
        }
    }
    reverse(path.begin(), path.end());
    return path; // size = E+1 if valid Euler tour
}

```

Input: $\text{adj}[u][i] = v, \text{id}$, $\text{adj}[v][j] = u, \text{id}$ if u and v adjacent (same id for both edges).

```

vi euler_undirected(int n, vector<vpi>& adj) {
    vi used, path; stack<int> st; st.push(0);
    int m=0; for (auto &v:adj) m += v.size(); used.assign(m/2,0);
    while(!st.empty()){
        int v=st.top();
        while(!adj[v].empty() && used[adj[v].back().second])
            adj[v].pop_back();
        if(adj[v].empty()){
            path.push_back(v);
            st.pop();
        } else {
            auto e=adj[v].back(); adj[v].pop_back();
            used[e.second]=1;
            st.push(e.first);
        }
    }
    reverse(path.begin(), path.end());
    return path;
}

```

2.7 Hungarian

Min cost assignment $O(n^3)$. Input cost matrix, output (min cost, assignment)

```

pair<int, vi> hungarian(const vvi &a) {
    int n = a.size(), m = a[0].size();
    vi u(n + 1), v(m + 1), p(m + 1), way(m + 1);
    for (int i = 1; i <= n; i++) {
        p[0] = i;
        vi minv(m + 1, 1e9); // INF constant
        vector<bool> used(m + 1, false);
        int j0 = 0;
        do {
            used[j0] = true;
            int i0 = p[j0], delta = 1e9, j1 = 0;
            for (int j = 1; j <= m; j++) if (!used[j]) {
                int cur = a[i0 - 1][j - 1] - u[i0] - v[j];
                if (cur < minv[j]) minv[j] = cur, way[j] = j0;
                if (minv[j] < delta) delta = minv[j], j1 = j;
            }
            for (int j = 0; j <= m; j++) {
                if (used[j]) u[p[j]] += delta, v[j] -= delta;
                else minv[j] -= delta;
            }
        } while (delta > 0);
    }
}

```

```

    }
    j0 = j1;
} while (p[j0] != 0);
do {
    int j1 = way[j0];
    p[j0] = p[j1];
    j0 = j1;
} while (j0);
}
vi match(n);
for (int j = 1; j <= m; j++) if (p[j]) match[p[j] - 1] = j - 1;
return {-v[0], match};
}

```

2.8 Iterative dfs

Iterative depth-first search

```

void dfs_iter(int start, vvi& adj, vi &output, vi& visited) {
    stack<int> st; st.push(start); visited[start] = 1;
    while (!st.empty()) {
        int v = st.top(); bool done = true;
        for (auto u : adj[v]) {
            if (!visited[u]) {
                visited[u] = 1; st.push(u);
                done = false; break;
            }
        }
        if (done) {
            st.pop(); output.push_back(v);
        }
    }
}

```

2.9 LCA

```

struct LCA {
    int n; vvi adj;
    vi depth, first, euler;
    RMQ rmq; // c.f. RMQ structure.
    LCA(int _n : n(_n), adj(n), depth(n), first(n), rmq(vi())) {}
    void add_edge(int u, int v) { adj[u].push_back(v); adj[v].push_back(u); }
    void dfs(int v, int p, int d) {
        first[v] = euler.size();
        depth[v] = d;
        euler.push_back(v);
        for (int u : adj[v]) if (u != p) {
            dfs(u, v, d+1);
            euler.push_back(v);
        }
    }
    void build(int root=0) {
        euler.clear();
        dfs(root, -1, 0);
        vi D(euler.size());
        for (int i=0; i<euler.size(); i++) D[i]=depth[euler[i]];
        rmq = RMQ(D);
    } // dist(a,b) = depth[a] + depth[b] - 2 * depth[lca];
    int query(int a, int b) {
        int l = first[a], r = first[b];
        if (l>r) swap(l,r);
        return euler[rmq.query(l,r+1)];
    }
};

```

2.10 Max Bipartite Matching

König's theorem: In a bipartite graph, the number of edges in a maximum matching equals the number of vertices in a minimum vertex cover.

Hopcroft–Karp $O(E\sqrt{V})$ left $[0..nL-1]$, right $[0..nR-1]$ adj from left to right.

```

int nL, nR;
vector<vi> adj;
vi dist, matchL, matchR;
bool bfs() {
    queue<int> q;
    dist.assign(nL, -1);
    for (int u = 0; u < nL; ++u)
        if (matchL[u] == -1) dist[u] = 0, q.push(u);
    bool found = 0;
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (int v : adj[u]) {
            int mu = matchR[v];
            if (mu == -1) found = 1;
            else if (dist[mu] == -1) dist[mu] = dist[u] + 1, q.push(mu);
        }
    }
    return found;
}
bool dfs(int u) {
    for (int v : adj[u]) {
        int mu = matchR[v];
        if (mu == -1 || (dist[mu] == dist[u] + 1 && dfs(mu))) {
            matchL[u] = v; matchR[v] = u;
            return 1;
        }
    }
    dist[u] = -1;
    return 0;
}
int hopcroftKarp() {
    matchL.assign(nL, -1);
    matchR.assign(nR, -1);
    int matching = 0;
    while (bfs())
        for (int u = 0; u < nL; ++u)
            if (matchL[u] == -1 && dfs(u))
                ++matching;
    return matching;
}

```

2.11 Max Flow Min Cut

Dinic's max flow $O(V^2E)$, $O(E\sqrt{V})$ for bipartite/unit. Edges from reachable nodes after flow form a min cut.

```

struct FlowEdge {
    int v, u;
    long long cap, flow = 0;
    FlowEdge(int v, int u, long long cap) : v(v), u(u), cap(cap) {}
};
struct Dinic {
    const long long flow_inf = 1e18;
    vector<FlowEdge> edges;
    vector<vi> adj;
    int n, m = 0;
    int s, t; // source, target
    vi level, ptr;
    queue<int> q;

```

```

Dinic(int n, int s, int t) : n(n), s(s), t(t) {
    adj.resize(n); level.resize(n); ptr.resize(n);
}

void add_edge(int v, int u, long long cap) {
    edges.emplace_back(v, u, cap);
    edges.emplace_back(u, v, 0);
    adj[v].push_back(m);
    adj[u].push_back(m + 1);
    m += 2;
}

bool bfs() {
    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (int id : adj[v]) {
            if (edges[id].cap == edges[id].flow) continue;
            if (level[edges[id].u] != -1) continue;
            level[edges[id].u] = level[v] + 1;
            q.push(edges[id].u);
        }
    }
    return level[t] != -1;
}

long long dfs(int v, long long pushed) {
    if (pushed == 0) return 0;
    if (v == t) return pushed;
    for (int& cid = ptr[v]; cid < (int)adj[v].size(); cid++) {
        int id = adj[v][cid]; int u = edges[id].u;
        if (level[v] + 1 != level[u]) continue;
        long long tr = dfs(u, min(pushed, edges[id].cap - edges[id].flow));
        if (tr == 0) continue;
        edges[id].flow += tr;
        edges[id ^ 1].flow -= tr;
        return tr;
    }
    return 0;
}

long long flow() {
    long long f = 0;
    while (true) {
        fill(level.begin(), level.end(), -1);
        level[s] = 0; q.push(s);
        if (!bfs()) break;
        fill(ptr.begin(), ptr.end(), 0);
        while (long long pushed = dfs(s, flow_inf))
            f += pushed;
    }
    return f;
}

vector<pii> min_cut_edges() {
    vector<bool> vis(n, false);
    queue<int> q;
    q.push(s); vis[s] = true;
    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (int id : adj[v]) {
            auto &e = edges[id];
            if (!vis[e.u] && e.cap > e.flow) {
                vis[e.u] = true;
                q.push(e.u);
            }
        }
    }
    vector<pii> cut;
    for (auto &e : edges) {

```

```

        if (vis[e.v] && !vis[e.u] && e.cap > 0) {
            cut.push_back({e.v, e.u});
        }
    }
    return cut;
};

};

2.12 Min Cost Flow
Finds min cost to send up to F flow (set F=INF for MCMF)  $O(FE \log V)$ . Handles neg costs (no neg cycles).

struct MCF {
    struct E { int v, rev, cap; ll cost; };
    int n; vector<vector<E>> g;
    MCF(int n): n(n), g(n) {}
    void addEdge(int u, int v, int cap, ll cost) {
        g[u].push_back({v, (int)g[v].size(), cap, cost});
        g[v].push_back({u, (int)g[u].size()-1, 0, -cost});
    }
    pair<ll, ll> minCostFlow(int s, int t, int maxf) {
        ll flow=0, cost=0; vll pot(n), dist(n); vi pv(n), pe(n);
        while (flow < maxf) {
            fill(dist.begin(), dist.end(), LLONG_MAX);
            priority_queue<pii, vector<pii>, greater<>> pq;
            dist[s]=0; pq.push({0, s});
            while (!pq.empty()) {
                auto [d, u] = pq.top(); pq.pop();
                if (d != dist[u]) continue;
                for (int i=0; i < (int)g[u].size(); i++) {
                    E &e = g[u][i]; if (!e.cap) continue;
                    ll nd = d + e.cost + pot[u] - pot[e.v];
                    if (nd < dist[e.v]) {
                        dist[e.v] = nd; pv[e.v] = u; pe[e.v] = i;
                        pq.push({nd, e.v});
                    }
                }
            }
            if (dist[t] == LLONG_MAX) break;
            for (int i=0; i < n; i++) if (dist[i] < LLONG_MAX) pot[i] += dist[i];
            int add = maxf - flow;
            for (int v=t; v != s; v = pv[v])
                add = min(add, g[pv[v]][pe[v]].cap);
            for (int v=t; v != s; v = pv[v]) {
                E &e = g[pv[v]][pe[v]];
                e.cap -= add; g[v][e.rev].cap += add;
                cost += add * e.cost;
            }
            flow += add;
        }
        return {flow, cost};
    }
};

};

```

2.13 Minimum Spanning Tree

Kruskal $O(E \log E)$ edges list of $\{u, v, \text{weight}\}$

```

int kruskal(int V, vvi &edges) {
    sort(edges.begin(), edges.end(), [](auto a, auto b) { return a[2] < b[2]; });
    union_find uf(V);
    int cost = 0, count = 0;
    for (auto &e : edges) {
        if (uf.find(e[0]) != uf.find(e[1])) {
            uf.unite(e[0], e[1]);

```

```

        cost += e[2];
        if (++count == V - 1) break;
    }
}
return cost;
}

```

Prim's $O((E + V) \log V)$ adj[u] list of (v,weight) adjacent to u.

```

int prim(int V, vector<vpi> &adj) {
    priority_queue<pii, vector<pii>, greater<pii>> q;
    vector<bool> vis(V, false);
    q.push({0, 0});
    int res = 0;
    while(!q.empty()){
        auto [w,u] = q.top();q.pop();
        if(vis[u])continue;
        res += w;
        vis[u] = true;
        for(auto v : adj[u]){
            if(!vis[v.first]){
                q.push({v.second, v.first});
            }
            //weight, vertex
        }
    }
    return res;
}
}

```

2.14 Shortest Path

Find shortest paths from src (no negative weights). $O((V+E)\log V)$

```

vi dijkstra(const vector<vector<pii>>& adj, int src) {
    vi dist(adj.size(), INT_MAX);
    priority_queue<pii, vector<pii>, greater<pii>> q;
    dist[src] = 0; q.push({0, src});
    while (!q.empty()) {
        auto [d, u] = q.top(); q.pop();
        if (d != dist[u]) continue;
        for (auto [v, w] : adj[u]) {
            if (d+w < dist[v]) {
                dist[v] = d+w;
                q.push({d+w, v});
            }
        }
    }
    return dist;
}
}

```

$O(VE)$ Shortest Path+neg edges; BFS from nodes with dist $-\infty$ for all neg-cycle reachable.

```

vi bellmanFord(int n, vvi& edges, int src) {
    vector<int> dist(n, INT_MAX);
    dist[src] = 0;
    for (int i = 0; i < n; i++) {
        for (auto edge : edges) {
            int u = edge[0];int v = edge[1];int wt = edge[2];
            if (dist[u] != INT_MAX && dist[u] + wt < dist[v]) {
                if(i == n - 1) return {-1};
                dist[v] = dist[u] + wt;
            }
        }
    }
    return dist;
}
}

```

All-pairs shortest paths (neg edges ok, no neg cycles) $O(V^3)$ $graph[i][i] = 0$, $graph[i][j] = w$ if edge $i \rightarrow j$ else INT_MAX

```

vector<vi> floydWarshall(vector<vi> graph) {
    int V = graph.size();
    auto dist = graph;
    for (int k = 0; k < V; ++k)
        for (int i = 0; i < V; ++i)
            for (int j = 0; j < V; ++j)
                if (dist[i][k] < INT_MAX && dist[k][j] < INT_MAX)
                    dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);

    return dist;
}

```

2.15 Strongly Connected Components

Kosaraju SCC $O(n+m)$

```

struct SCC {
    int n, ordn = 0, scc_cnt = 0;
    vvi adj, adjt; vi vis, ord, scc;
    SCC(int n): n(n), adj(n), adjt(n), vis(n,0), ord(n), scc(n) {}
    void add_edge(int u,int v){ adj[u].push_back(v); adjt[v].push_back(u); }
    void dfs(int u){
        vis[u]=1;
        for(int v:adj[u]) if(!vis[v]) dfs(v);
        ord[ordn++]=u;
    }
    void dfst(int u){
        scc[u]=scc_cnt; vis[u]=0; // SCCs IDs are in reverse topological order.
        for(int v:adjt[u]) if(vis[v]) dfst(v);
    }
    void kosaraju(){
        for(int i=0;i<n;i++) if(!vis[i]) dfs(i);
        for(int i=ordn-1;i>=0;i--) if(vis[ord[i]]){ scc_cnt++; dfst(ord[i]); }
    }
};

```

2.16 Toposort

TopoSort via DFS $O(V + E)$.

```

void dfs(int u, vector<vi> &adj, vi &vis, vi &res) {
    vis[u] = 1;
    for (int v : adj[u])
        if (!vis[v])
            dfs(v, adj, vis, res);
    res.push_back(u);
}
vi toposort(vector<vi> &adj) {
    int n = adj.size();
    vi vis(n, 0), res;
    for (int i = 0; i < n; i++)
        if (!vis[i]) dfs(i, adj, vis, res);
    reverse(res.begin(), res.end());
    return res;
}

```

2.17 Useful theory

3 Dynamic Programming

3.1 Coin Change

Given coins $c_1, \dots, c_k$ and a sum S :

Min coins to make S : DP Formula: $dp[x]$: min coins to make x , $dp[0] = 0$. $dp[x] = \min_{c_i \leq x} (dp[x - c_i] + 1)$. Ans = $dp[S]$.

Number of ways to make S : DP formula: $dp[x]$: #ways to make x , $dp[0] = 1$. $dp[x] =$

$\sum_{c_i \leq x} dp[x - c_i]$. Ans = dp[S].

Number of ways to make S, $O(n \times \text{sum})$

```
def count(coins, s):
    n = len(coins) ; dp = [0] * (s + 1) ; dp[0] = 1
    for i in range(n):
        for j in range(coins[i], s + 1):
            dp[j] += dp[j - coins[i]]
    return dp[s]
```

3.2 Convex Hull Trick

Maintains lines $y = m \cdot x + b$, supports add(m,b) and query(x) for max y at given x. Works in $O(\log n)$ per operation using a multiset and integer intersections.

```
struct Line {
    mutable ll m, b, p;
    bool operator<(const Line& o) const { return m < o.m; }
    bool operator<(ll x) const { return p < x; }
};

struct LineContainer : multiset<Line, less<>> {
    const ll inf = LLONG_MAX;
    ll div(ll a, ll b) {
        return a / b - ((a ^ b) < 0 && a % b);
    }
    bool isect(iterator x, iterator y) {
        if (y == end()) { x->p = inf; return false; }
        if (x->m == y->m) x->p = x->b > y->b ? inf : -inf;
        else x->p = div(y->b - x->b, x->m - y->m);
        return x->p >= y->p;
    }
    void add(ll m, ll b) {
        auto z = insert({m, b, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p) isect(x, erase(y));
    }
    ll query(ll x) {
        assert(!empty());
        auto l = *lower_bound(x);
        return l.m * x + l.b;
    }
};
```

3.3 Kadane

Given an array $a_1, \dots, a_n$, find $\max_{1 \leq i \leq j \leq n} \sum_{k=i}^j a_k$

DP formula: dp_i : max sum of subarray ending at i , $dp_i = \max(a_i, dp_{i-1} + a_i)$, Ans = $\max_i dp_i$.

Kadane 1D $O(n)$

```
def kadane1D(arr):
    maxSum = arr[0] ; DP = arr[0]
    for i in range(1, len(arr)):
        DP = max(DP + arr[i], arr[i])
        maxSum = max(maxSum, DP)
    return maxSum
```

2D-version : For $r_1 \leq r_2$: $\text{colSum}[c] = \sum_{r=r_1}^{r_2} A_{r,c}$, $\text{maxSum} = \max(\text{maxSum}, \text{Kadane1D}(\text{colSum}))$.

Kadane 2D $O(n^3)$

```
def kadane2D(mat):
    rows = len(mat) ; cols = len(mat[0]) ; maxSum = float('-inf') ; temp = [0] * rows
    for left in range(cols):
        temp = [0] * rows
        for right in range(left, cols):
            for row in range(rows):
                temp[row] += mat[row][right]
```

```
sumValue = kadane1D(temp)
maxSum = max(maxSum, sumValue)
return maxSum
```

3.4 Knapsack

n items, each with value v_i and weight w_i . Max total weight W . DP formula: $dp[i][j]$ max value using i items, where max total weight is j . $dp[i][w] = \max(dp[i-1][w], dp[i-1][w - w_i] + v_i)$.

Space optimized version $O(n \times W)$

```
def knapsack(W, val, wt):
    dp = [0] * (W + 1)
    for i in range(1, len(wt) + 1):
        for j in range(W, wt[i - 1] - 1, -1):
            dp[j] = max(dp[j], dp[j - wt[i - 1]] + val[i - 1])
    return dp[W]
```

3.5 Longest Increasing Subsequence

ℓ_i : LIS length ending at i . DP formula : $\ell_i = 1 + \max_{1 \leq j < i \atop a_j < a_i} \ell_j$ (or 1 if none). Ans = $\max_i \ell_i$.

Predecessor $p_i = \arg \max_{1 \leq j < i \atop a_j < a_i} \ell_j$.

$O(n^2)$

```
def LIS(arr):
    n = len(arr) ; lis = [1] * n
    for i in range(1, n):
        for prev in range(0, i):
            if arr[i] > arr[prev]:
                lis[i] = max(lis[i], lis[prev] + 1)
    return max(lis)
```

Note that $O(n \log n)$ is possible with BS

```
def LIS_BS(arr):
    n = len(arr) ; ans = [] ; ans.append(arr[0])
    for i in range(1, n):
        if arr[i] > ans[-1]:
            ans.append(arr[i])
        else:
            low = 0
            high = len(ans) - 1
            while low < high:
                mid = low + (high - low) // 2
                if ans[mid] < arr[i]:
                    low = mid + 1
            else:
                high = mid
            ans[low] = arr[i]
    return len(ans)
```

3.6 MinPathSum

Given a grid with non negative ints, where you can only move right/down finds the min sum path from top-left to bottom-right.

$O(n \times m)$ Modifies grid.

```
ll minPathSum(vvi &grid){
    for (int i = 0; i < grid.size(); i++){
        for (int j = 0; j < grid[0].size(); j++){
            if (i == 0 && j == 0) continue;
            if (i == 0) grid[i][j] += grid[i][j - 1];
            else if (j == 0) grid[i][j] += grid[i - 1][j];
            else grid[i][j] += min(grid[i - 1][j], grid[i][j - 1]);
        }
    }
    return grid[grid.size() - 1][grid[0].size() - 1];
}
```

4 Strings

Finds the occurrences of a set of patterns in a text simultaneously $O(n + m + matches)$, where $n = \text{len}(\text{text})$, $m = \text{total pattern length}$.

```
def search_text(text, keywords):
```

Traveling Salesman Problem $O(n^{2^{2n}})$ DP formula: $\text{dp}[S][i]$: min cost to start at 0, visit all $v \in S$, end at i . $\text{dp}[\{0\}][0] = 0$. $\text{dp}[S][i] = \min_{j \in S \setminus \{i\}} \text{dp}[S \setminus \{i\}][j] + w_{j,i}$. Ans = $\min_i \text{dp}[V][i] + w_{i,0}$.

}

```
return res
```

Search for a pattern within a text in $O(n + m)$, where n is the size of the text and m the length of the pattern. Length of the longest proper prefix which is also a suffix of the substring ending at that position.

```
return res
```

Longest palindromic substring $O(n)$. Max of P corresponds to len(LPS)

```
if i + P[i] > R: # update center and right boundary
```

C, R = i, i + P[i]

4.4 SuffixArray

SuffixArray i.e. starting positions of suffixes in lexicographic order. $O(n \log^2 n)$

```
def build_sa(s):
    n = len(s)
    sa = list(range(n))
    rank = [ord(ch) for ch in s]    # initial ranks by 1 char
    k = 1
    while k < n: # sort by first k chars
        sa.sort(key=lambda x: (rank[x], rank[x+k] if x+k < n else -1))
        temp = [0]*n
        temp[sa[0]] = 0
        for i in range(1, n):
            prev = sa[i-1]
            cur = sa[i]
            prev_key = (rank[prev], rank[prev+k] if prev+k < n else -1)
            cur_key = (rank[cur], rank[cur+k] if cur+k < n else -1)
            temp[cur] = temp[prev] + (cur_key > prev_key)
        rank = temp
        if rank[sa[-1]] == n-1: break # all ranks distinct
        k <= 1
    return sa
```

Longest Common Prefix between suffixes sa[i] and sa[i+1] $O(n)$

```
def build_lcp(s, sa):
    n = len(s)
    rank = [0]*n
    for i in range(n): rank[sa[i]] = i
    lcp = [0]*n
    k = 0
    for i in range(n):
        if rank[i] == n-1: k = 0; continue
        j = sa[rank[i] + 1]
        while i+k < n and j+k < n and s[i+k] == s[j+k]:
            k += 1
        lcp[rank[i]] = k
        if k: k -= 1
    return lcp
```

Longest Repeated Substring = max LCP over all adjacent suffixes.

```
def lrs(s):
    sa = build_sa(s)
    lcp = build_lcp(s, sa)
    idx = lcp.index(max(lcp))
    return (lcp[idx], idx) # (length, position in SA)
```

Finds all occurrences of pat in s via binary search on the SA. Returns [L, R]: all SA positions where suffix starts with pat.

```
def search(s, pat, sa):
    n, m = len(s), len(pat)
    l, r = 0, n-1
    while l <= r:
        mid = (l+r)//2
        if s[sa[mid]:sa[mid]+m] < pat: l = mid+1
        else: r = mid-1
    start = l
    l, r = 0, n-1
    while l <= r:
        mid = (l+r)//2
        if s[sa[mid]:sa[mid]+m] <= pat: l = mid+1
        else: r = mid-1
    return (start, r) if start <= r else (-1, -1)
```

Longest Common Substring between two strings. Build SA/LCP on s1#s2\$, then the LCS occurs between adjacent suffixes coming from different strings.

```
def lcs(s1, s2):
    s = s1 + '#' + s2 + '$'
    sa = build_sa(s)
    lcp = build_lcp(s, sa)
    idx, max_len = 0, 0
    for i in range(1, len(s)):
        a = sa[i] < len(s1)
        b = sa[i-1] < len(s1)
        if a != b and lcp[i] > max_len:
            max_len, idx = lcp[i], i
    return (max_len, idx)
```

5 Arrays

5.1 Inversions

Count pairs where order flips between arrays. $O(n \log n)$

```
ll inversions(vi& a, vi& b) {
    int n = a.size();
    unordered_map<int,int> pos;
    for (int i = 0; i < n; i++) pos[b[i]] = i + 1;
    Fenwick t(n); // C.f. Fenwick tree
    ll inv = 0;
    for (int i = 0; i < n; i++) {
        inv += i - t.sum(pos[a[i]]);
        t.add(pos[a[i]], 1);
    }
    return inv;
}
```

6 Math

6.1 Fourier Transform

FFT: in-place, invert=false for forward, true for inverse Input size must be power of two $O(n \log n)$

```
void fft(vector<complex<double>> &a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1)
            j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        double ang = 2 * M_PI / len * (invert ? -1 : 1);
        complex<double> wlen(cos(ang), sin(ang));
        for (int i = 0; i < n; i += len) {
            complex<double> w(1);
            for (int j = 0; j < len / 2; j++) {
                complex<double> u = a[i + j];
                complex<double> v = a[i + j + len / 2] * w;
                a[i + j] = u + v;
                a[i + j + len / 2] = u - v;
                w *= wlen;
            }
        }
    }
    if (invert) {
        for (auto &x : a) x /= n;
    }
}
```

```

void multiply(vi &a, vi &b) {
    vector<complex<double>> fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1; while (n < a.size() + b.size()) n <= 1; // next power of 2
    fa.resize(n); fb.resize(n);
    fft(fa, false); fft(fb, false); // forward FFT
    for (int i = 0; i < n; i++) fa[i] *= fb[i]; // pointwise multiply
    fft(fa, true); // inverse FFT
    vi res(n);
    for (int i = 0; i < n; i++) res[i] = round(fa[i].real());
    return res;
}

```

Solve $Ax = b$ using Gaussian elimination $O(n^3)$

```
n = len(a)
a = [row[:] + [b[i]] for i, row in enumerate(a)]
for i in range(n):
    pivot_row = max(range(i, n), key=lambda r: abs(a[r][i]))
    if abs(a[pivot_row][i]) < 1e-12:
        return None
    a[i], a[pivot_row] = a[pivot_row], a[i]
    pivot = a[i][i]
    for j in range(i, n + 1):
        a[i][j] /= pivot
    for r in range(i + 1, n):
        factor = a[r][i]
        for c in range(i, n + 1):
            a[r][c] -= factor * a[i][c]
x = [0] * n
for i in range(n - 1, -1, -1):
    x[i] = a[i][n]
    for j in range(i + 1, n):
        x[i] -= a[i][j] * x[j]
return x
```

```
def simplex(A, b, c):
    m, n = len(A), len(c)
    tableau = []
    for i in range(m):
        tableau.append(A[i][:] + [1 if j == i else 0 for j in range(m)] + [b[i]])
    tableau.append([-c_j for c_j in c] + [0] * (m + 1))
    basic = list(range(n, n + m))
    while True:
        entering = -1
        for j in range(n + m):
            if tableau[-1][j] < -1e-12:
                if entering == -1 or tableau[-1][j] < tableau[-1][entering]:
                    entering = j
        if entering == -1: # Optimal solution found
            break
        leaving = -1
        min_ratio = float('inf')
        for i in range(m):
            if tableau[i][entering] > 1e-12:
                ratio = tableau[i][-1] / tableau[i][entering]
                if ratio < min_ratio - 1e-12:
                    min_ratio = ratio
                    leaving = i
        if leaving == -1: # Unbounded problem
```

6.3 Matrix Operations

Calculates a^p in $O(n^3 \log p)$. Watch out for overflow. Modifies matrix.

Gauss-Jordan on an augmented matrix, returns determinant, modifies matrix. $O(n^3)$

```
double gaussJordan(matrix &a, int &rank, int &augRank) {
    int n = a.size(), m = a[0].size() - 1; // n eqs, m vars
    double det = 1.0; rank = 0;
    for (int j = 0; j <= m; j++) {
        int pivotRow = -1;
        double maxVal = 0;
        for (int i = rank; i < n; i++)
            if (pivotRow < 0 || abs(a[i][j]) > maxVal) {
                maxVal = abs(a[i][j]);
                pivotRow = i;
            }
        if (maxVal > 1e-7) {
            double c = a[pivotRow][j];
            if (pivotRow != rank) det = -det;
            det *= c;
            for (int k = j; k <= m; k++) {
                swap(a[pivotRow][k], a[rank][k]);
                a[rank][k] /= c;
            }
            for (int i = rank + 1; i < n; i++) {
                c = a[i][j];

```

```

        for (int k = j; k <= m; k++)
            a[i][k] -= c * a[rank][k];
    }
    if (j < m) rank++;
}
augRank = rank;
for (int i = rank; i < n; i++)
    if (abs(a[i][m]) > 1e-6) { augRank++; break; }
if (rank != n) det = 0;
return det;
}

```

Solve $Ax=b$. Input: $[A|b]$ (augmented matrix) $O(n^3)$. Modifies Matrix.

```

void solve(matrix &a, vi &solution) {
    int n = a.size(), m = a[0].size() - 1;
    solution.assign(m, 0.0);
    for (int i = n - 1; i >= 0; i--) {
        double c = a[i][m];
        for (int j = i + 1; j < m; j++)
            c -= a[i][j] * solution[j];
        solution[i] = c;
    }
}

```

6.4 Number Theory

Linear diophantine equations: Let $a, b, c \in \mathbb{N}$. The equation $ax + by = c$ has integer solutions iff c is a multiple of $\gcd(a, b)$. If (x_0, y_0) is a solution for the equation (c.f. Extended Euclidean Algorithm), then all integer solutions are given by $(x_0 - b't, y_0 + a't)$ with $t \in \mathbb{N}, a' = \frac{a}{\gcd(a, b)}, b' = \frac{b}{\gcd(a, b)}$.

Chinese remainder theorem: Let $n_1, \dots, n_k \in \mathbb{N}$ pairwise coprime, and $b_1, \dots, b_k \in \mathbb{N}$ The system of linear congruencies $x \equiv b_i \pmod{n_i}$ for $1 \leq i \leq k$ has a unique solution $\pmod{\prod_{i=1}^k n_i}$, constructed as follows: Let $a_i = \frac{N}{n_i}$ and c_i its inverse $\pmod{n_i}$. Then, $x \equiv \sum_{i=1}^k a_i b_i c_i \pmod{\prod_{i=1}^k n_i}$

$\gcd(a, b) = \gcd(a, b \bmod a)$, $\text{lcm}(a, b) = \frac{ab}{\gcd(a, b)}$, $\gcd(a, b, c) = \gcd(\gcd(a, b), c)$, $\text{lcm}(a, b, c) = \text{lcm}(\text{lcm}(a, b), c)$

6.5 Useful NT Functions in Python

Sieve of Gries-Misra $O(n)$. Produces prime list and SPF array

```

def gries_misra(n):
    primes = []
    factor = [0] * n
    for x in range(2, n):
        if not factor[x]:
            # no factor found
            factor[x] = x
            # meaning x is prime
            primes.append(x)
        for p in primes:
            # loop over primes found so far
            if p > factor[x] or p * x >= n:
                break
            factor[p * x] = p
            # p is the smallest factor of p * x
    return primes, factor

```

Returns (u,v) s.t. $au + bv = \gcd(a, b)$ in $O(\log a + \log b)$

```

def bezout(a, b):
    if b == 0:
        return (1, 0)
    u, v = bezout(b, a % b)
    return (v, u - (a // b) * v)

```

Inverse of a in $\mathbb{Z}_p$ in $O(\log a + \log p)$

```

def inv(a, p):
    return bezout(a, p)[0] % p

```

$\binom{n}{k} \bmod p$ in $O(k)$

```

def binom_modulo(n, k, p):
    prod = 1
    for i in range(k):
        prod = (prod * (n - i) * inv(i + 1, p)) % p
    return prod

```

Constructs an arithmetic expression tree. Input: tokens e.g. ["1", "+", "2"]

```

def arithm_expr_parse(tokens):
    vals, ops = [], []
    PRIO = {'+':1, '-':1, '*':2, '/':2, '(':0}
    for t in tokens:
        if t in PRIO:
            while ops and PRIO[ops[-1]] >= PRIO[t] and t != '(':
                vals.append((vals.pop(-2), ops.pop(), vals.pop()))
                ops.append(t)
            elif t == ')':
                while ops[-1] != '(':
                    vals.append((vals.pop(-2), ops.pop(), vals.pop()))
                    ops.pop()
                else:
                    vals.append(int(t) if t.isdigit() else t)
        while ops:
            vals.append((vals.pop(-2), ops.pop(), vals.pop()))
    return vals[0]

```

6.6 Useful NT Functions in c++

Sieve of Eratosthenes: $O(n \log \log n)$ sieve[i] = 0 if i prime, spf otherwise for fast fact

```

vi sieve(int n){
    vi sieve(n,0);
    for(int i = 2; i*i < n ; i++)
        if(!sieve[i])
            for(int j = i*i ; j < n ; j += i)
                if(!sieve[j]) sieve[j] = i;
    return sieve;
}

```

Fast exponentiation $O(\log b)$: fast $a^b \bmod p$.

```

int modular_exp(int a, int b, int p){
    int res = 1;
    while(b > 0){
        if(b & 1) res = (1LL * a * res) % p;
        b = b >> 1;
        a = (1LL * a * a) % p;
    }
    return res;
}

```

Euler's φ : Number of ints $\leq n$ coprime to n .

```

int phi(int n, vi& primes) {
    int res = n;
    for (int p : primes) {
        if (1LL * p * p > n) break;
        if (n % p == 0) {
            while (n % p == 0) n /= p;
            res -= res / p;
        }
    }
    if (n > 1) res -= res / n;
    return res;
}

```

Extended Euclidean Algorithm Solves $ax + by = \gcd(a, b)$.

```

11 eea(11 a, 11 b, 11& x, 11& y) {
    11 xx = y = 0, yy = x = 1;
    while (b) {
        11 q = a / b, t = b; b = a%b; a = t;
        t = xx; xx = x - q*xx; x = t;
        t = yy; yy = y - q*yy; y = t;
    }
    return a; // Returns gcd(a,b)
}

```

7 Geometry

7.1 Circles

Retourne par reference le centre d'un cercle qui passe par 2 points et a un rayon donne

```

bool circle2PtsRad(point p1, point p2, double r, point &c) {
    double d2 = (p1.x - p2.x) * (p1.x - p2.x) +
                (p1.y - p2.y) * (p1.y - p2.y);
    double det = r * r / d2 - 0.25;
    if (det < 0.0) return false;
    double h = sqrt(det);
    c.x = (p1.x + p2.x) * 0.5 + (p1.y - p2.y) * h;
    c.y = (p1.y + p2.y) * 0.5 + (p2.x - p1.x) * h;
    return true;
}

```

Rayon du cercle inscrit donnees les longueurs des cotes

```

double rInCircle(double ab, double bc, double ca) {
    return heron(ab, bc, ca) / (0.5 * (ab+ bc+ca)); }

```

Retourne le centre et le rayon du cercle inscrit d'un triangle (par reference)

```

bool inCircle(point p1, point p2, point p3, point &ctr, double &r) {
    r = rInCircle(p1.dist(), p2.dist(), p3.dist());
    if (abs(r) < EPS) return false; // no inCircle center
    line l1, l2; // compute these two angle bisectors
    double ratio = dist(p1, p2) / dist(p1, p3);
    point p = translate(p2, scale(toVec(p2, p3), ratio / (1 + ratio)));
    pointsToLine(p1, p, l1);
    ratio = dist(p2, p1) / dist(p2, p3);
    p = translate(p1, scale(toVec(p1, p3), ratio / (1 + ratio)));
    pointsToLine(p2, p, l2);
    areIntersect(l1, l2, ctr); // get their intersection point
    return true;
}

```

Rayon du cercle circonscrit donnees les longueurs des cotes

```

double rCircumCircle(double ab, double bc, double ca) {
    return ab * bc * ca / (4.0 * heron(ab, bc, ca)); }

```

Retourne le centre et le rayon du cercle circonscrit d'un triangle (par reference)

```

bool circumCircle(point p1, point p2, point p3, point &ctr, double &r){
    double a = p2.x - p1.x, b = p2.y - p1.y, c = p3.x - p1.x, d = p3.y - p1.y;
    double e = a * (p1.x + p2.x) + b * (p1.y + p2.y);
    double f = c * (p1.x + p3.x) + d * (p1.y + p3.y);
    double g = 2.0 * (a * (p3.y - p2.y) - b * (p3.x - p2.x));
    if (fabs(g) < EPS) return false;
    ctr.x = (d*e - b*f) / g; ctr.y = (a*f - c*e) / g;
    r = dist(p1, ctr); // r = distance from center to 1 of the 3 points
    return true;
}

```

Retourne l'orthocentre d'un triangle. r n'a pas de sens ici

```

bool orthocenter(point a, point b, point c, point & ctr, double &r){
    return circumCircle(a+b-c,b+c-a,c+a-b, ctr, r);
}

```

7.2 Closest Points

Minimum Euclidean distance between two distinct points. $O(n \log n)$

```

vector<point> p;
double cp(int l, int r) {
    if (l >= r) return DBL_MAX;
    int mid = (l + r) / 2;
    double d = min(cp(l, mid), cp(mid + 1, r));
    vector<point> strip;
    for (int i = l; i <= r; i++)
        if (fabs(p[i].x - p[mid].x) <= d)
            strip.push_back(p[i]);
    sort(strip.begin(), strip.end(), [](point a, point b){return a.y < b.y;});
    for (int i = 0; i < strip.size(); i++)
        for (int j = i+1; j < strip.size() && (strip[j].y - strip[i].y) <= d; j++)
            d = min(d, dist(strip[i], strip[j]));
    return d;
}

```

7.3 Points and Lines

#define EPS 1e-6

Point template, use long long or double, avoid int.

```

template <class T>
struct Point {
    T x, y;
    explicit Point(T x=0, T y=0) : x(x), y(y) {}
    bool operator<(P p) const { return tie(x,y) < tie(p.x,p.y); }
    bool operator==(P p) const { return tie(x,y)==tie(p.x,p.y); }
    P operator+(P p) const { return P(x+p.x, y+p.y); }
    P operator-(P p) const { return P(x-p.x, y-p.y); }
    P operator*(T d) const { return P(x*d, y*d); }
    P operator/(T d) const { return P(x/d, y/d); }
    T dot(P p) const { return x*p.x + y*p.y; }
    T cross(P p) const { return x*p.y - y*p.x; }
    T cross(P a, P b) const { return (a-*this).cross(b-*this); }
    T dist2() const { return x*x + y*y; }
    double dist() const { return sqrt((double)dist2()); }
}

```

angle to x-axis in interval [-pi, pi]

```

    double angle() const { return atan2(y, x); }
    P unit() const { return *this/dist(); } // makes dist()=1
    P perp() const { return P(-y, x); } // rotates +90 degrees
    P normal() const { return perp().unit(); }
}

```

returns point rotated 'a' radians ccw around the origin

```

    P rotate(double a) const {
        return P(x*cos(a)-y*sin(a),x*sin(a)+y*cos(a)); }
};

```

Vector $a \cdot b$

```

point toVec(point a, point b) { return point(b.x-a.x, b.y-a.y); }
point scale(point v, double s) { return point(v.x*s, v.y*s);}
point translate(point p, point v) { return point(p.x + v.x, p.y + v.y);}
double cross(point a, point b, point c) { return toVec(a, b).cross(toVec(b, c)); }
double dist(point p1, point p2) { return hypot(p1.x - p2.x, p1.y - p2.y);}
double distSq(point p1, point p2) {
    return (p1.x - p2.x)*(p1.x - p2.x)+ (p1.y - p2.y)*(p1.y - p2.y);
}

```

pour accepter des points colineaires remplacer $\cdot$ par $\cdot =$ et ajouter un EPS peut-etre returns true if point r is on the left side of line pq


```

bool ccw(point p, point q, point r) {
    return (toVec(p, q).cross(toVec(p, r))) > 0;
}

bool collinear(point p, point q, point r) {
    return abs(toVec(p, q).cross(toVec(p, r))) < EPS;
}

int insideCircle(point p, point c, double r) {
    double eucSq = distSq(p, c);
    double rSq = r*r;
    if (fabs(eucSq - rSq) > EPS && eucSq < rSq) return 0; // inside
    else if (fabs(eucSq - rSq) < EPS) return 1; // border
    else return 2; // outside
}

```

returns $\angle AOB$ in rad

```

double angle(point a, point o, point b) {
    point oa = toVec(o, a), ob = toVec(o, b);
    return acos(oa.dot(ob) / sqrt(oa.dist2() * ob.dist2()));
}

```

distance from p to the line defined by two points a and b. The closest point to p on the segment is stored in c.

```

double distToLine(point p, point a, point b, point &c) {
    point ap = toVec(a, p), ab = toVec(a, b);
    double u = ap.dot(ab) / ab.dist2();
    c = translate(a, scale(ab, u));
    return dist(p, c);
}

```

distance from p to the segment defined by two points a and b. The closest point to p on the segment is stored in c.

```

double distToSegment(point p, point a, point b, point &c) {
    auto ap = toVec(a, p), ab = toVec(a, b);
    double u = ap.dot(ab) / ab.dist2();
    if (u < 0.0) {
        c = point(a.x, a.y);
        return ap.dist();
    } else if (u > 1.0) {
        c = point(b.x, b.y);
        return toVec(p, b).dist();
    } else return distToLine(p, a, b, c);
}

```

Line representation using equation $ax + by + c = 0$

```

struct line { double a, b, c; };

```

Line that passes through two points, stored in the third parameter.

```

void pointsToLine(point p1, point p2, line &l) {
    if (fabs(p1.x - p2.x) < EPS) { // vertical
        l.a = 1.0;   l.b = 0.0;   l.c = -p1.x;
    } else {
        l.a = -(double)(p1.y - p2.y) / (p1.x - p2.x); l.b = 1.0;
        l.c = -(double)(l.a * p1.x) - p1.y;
    }
}

```

```

bool areParallel(line l1, line l2) { // check coefficients a & b
    return (fabs(l1.a-l2.a) < EPS) && (fabs(l1.b-l2.b) < EPS); }

```

Solves the linear system to find the intersection of two lines

```

bool areIntersect(line l1, line l2, point &p) {
    if (areParallel(l1, l2)) return false; // no intersection
    p.x = (l2.b * l1.c - l1.b * l2.c) / (l2.a * l1.b - l1.a * l2.b);
    if (fabs(l1.b) > EPS) p.y = -(l1.a * p.x + l1.c);
    else p.y = -(l2.a * p.x + l2.c);
    return true;
}

```

7.4 Polygons

```

#define PI 3.14159265358979323

```

Convex Hull

```

vector<point> andrew(vector<point> &P) {
    int n = P.size(), k = 0;
    vector<point> H(2*n);
    sort(P.begin(), P.end());
}

```

Build lower hull

```

    for (int i = 0; i < n; i++) {
        while (k >= 2 && !ccw(H[k-2], H[k-1], P[i])) k--;
        H[k++] = P[i];
    }

```

Build upper hull

```

    for (int i = n-2, t = k+1; i >= 0; i--) {
        while (k >= t && !ccw(H[k-2], H[k-1], P[i])) k--;
        H[k++] = P[i];
    }
    H.resize(k);
    return H;
}

```

```

double heron(double a, double b, double c){
    double s = (a+b+c)/2;
    return sqrt(s*(s-a)*(s-b)*(s-c));
}

```

```

double area(const vector<point> &P) {
    double result = 0.0, x1, y1, x2, y2;
    for (int i = 0; i < (int)P.size()-1; i++)
        result += (P[i].x * P[i+1].y - P[i+1].x * P[i].y);
    return abs(result) / 2.0; }

```

returns true if point p is in either convex/concave polygon P

```

bool inPolygon(point pt, const vector<point> &P) {
    if ((int)P.size() == 0) return false;
    double sum = 0;
    for (int i = 0; i < (int)P.size()-1; i++) {
        if (ccw(pt, P[i], P[i+1]))
            sum += angle(P[i], pt, P[i+1]);
        else sum -= angle(P[i], pt, P[i+1]); }
    return fabs(fabs(sum) - 2*PI) < EPS; }

```

$O(\log n)$ Seulement pour les convexes (ordonnes ccw)!

```

bool inConvexPolygon(point q, vector<point> const &p) {
    if (cross(p[0], p[1], q)<0 || cross(p[p.size()-2], p[0], q)<0)
        return false;
    int ini = 1, fin = p.size() - 2, mid;
    while (ini != fin - 1) {
        mid = (ini + fin) / 2;
        if (cross(p[0], p[mid], q) < 0) fin = mid;
        else ini = mid;
    }
    if (cross(p[ini], p[fin], q)<0) return false;
    return true;
}

```

Cut Polygon returns the point on segment PQ that is closest to line AB

```

point lineIntersectSeg(point p, point q, point A, point B) {
    double a = B.y - A.y, b = A.x - B.x, c = B.x * A.y - A.x * B.y;
    double u = abs(a * p.x + b * p.y + c), v = abs(a * q.x + b * q.y + c);
    return point((p.x * v + q.x * u) / (u+v), (p.y * v + q.y * u) / (u+v)); }

```

Cuts a polygon, keeping only the points on the left side of the directed line $a \rightarrow b$

```
vector<point> cutPolygon(point a, point b, const vector<point> &Q) {
    vector<point> P;
    for (int i = 0; i < (int)Q.size(); i++) {
        double left1 = toVec(a, b).cross(toVec(a, Q[i])), left2 = 0;
        if (i != (int)Q.size()-1) left2 = toVec(a, b).cross(toVec(a, Q[i+1]));
        if (left1 > -EPS) P.push_back(Q[i]); // Q[i] is on the left of ab
        if (left1 * left2 < -EPS) // edge (Q[i], Q[i+1]) crosses line ab
            P.push_back(lineIntersectSeg(Q[i], Q[i+1], a, b));
    }
    if (!P.empty() && !(P.back() == P.front()))
        P.push_back(P.front()); // make P's first point = P's last point
    return P; }
```

7.5 Useful Theory

Triangles Triangle dont les côtés sont a, b, c , les hauteurs sont h_a, h_b, h_c , les médianes sont m_a, m_b, m_c , le demi-périmètre est $s = \frac{a+b+c}{2}$ et l'aire est A . $\eta = \frac{1}{4R}$

Heron-like area formulae

$$\begin{aligned} A &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \left(4\sqrt{\eta(\eta-h_a^{-1})(\eta-h_b^{-1})(\eta-h_c^{-1})}\right)^{-1} \\ &= \frac{4}{3}\sqrt{s(s-m_a)(s-m_b)(s-m_c)} \end{aligned}$$

Rayon du cercle circonscrit $R = \frac{abc}{4A}$

Rayon du cercle inscrit $r = \frac{A}{s}$

Médianes divisent en deux sous-triangles de même aire.

$$m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

Bissectrices divisent les angles en deux parties égales.

$$b_a = \sqrt{bc\left(1 - \left(\frac{a}{b+c}\right)^2\right)}$$

Loi des sinus $\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c} = \frac{1}{2R}$

Loi des cosinus $a^2 = b^2 + c^2 - 2bc \cos \alpha$

Théorème de la tangente $\frac{a+b}{a-b} = \frac{\tan \frac{\alpha+\beta}{2}}{\tan \frac{\alpha-\beta}{2}}$

Quadrilatères

Quadrilatères cycliques (inscriptibles) Soit un quadrilatère **cyclique** de côtés a, b, c, d et diagonales e, f . Alors $ef = ac + bd$ et l'aire A peut être calculée grâce à la

Formule de Brahmagupta

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

où $s = \frac{a+b+c+d}{2}$

Coordonnées sphériques

$$\begin{aligned} x &= r \sin \theta \cos \phi & r &= \sqrt{x^2 + y^2 + z^2} \\ y &= r \sin \theta \sin \phi & \theta &= \arccos(z/\sqrt{x^2 + y^2 + z^2}) \\ z &= r \cos \theta & \phi &= \operatorname{atan2}(y, x) \end{aligned}$$

Centre de gravité: Soit A un sous-ensemble mesurable borné de $\mathbb{R}^d$, de mesure positive. Le centre de gravité de A est le point $G = (g_1, \dots, g_d) \in \mathbb{R}^d$ dont les coordonnées sont définies par:

$$g_i = \frac{1}{\lambda(A)} \int_A x_i dx_1 \dots dx_d$$

8 Utilities

8.1 C++ to Python

Initialization / Fixed size

```
vector<int> v(n, 0); // size n, all 0
vector<vector<int>> g(n, vector<int>(m, 0));
```

```
v = [0]*n
g = [[0]*m for _ in range(n)] # avoid [[0]*m]*n
```

Push / Pop / Insert / Erase

```
v.push_back(x);
v.pop_back();
v.insert(v.begin()+i, x);
v.erase(v.begin()+i);
v.erase(remove(v.begin(), v.end(), x), v.end()); // remove all x
```

```
v.append(x)
v.pop()
v.insert(i, x)
del v[i]
v = [e for e in v if e != x] # remove all x
```

Sorting / Unique / Reverse

```
sort(v.begin(), v.end());
sort(v.rbegin(), v.rend());
v.erase(unique(v.begin(), v.end()), v.end());
reverse(v.begin(), v.end());
```

```
v.sort()
v.sort(reverse=True)
v = sorted(set(v)) # deduplicate
v.reverse()
```

Binary search / Search

```
auto it = lower_bound(v.begin(), v.end(), x);
bool exists = (find(v.begin(), v.end(), x) != v.end());
```

```
import bisect
i = bisect.bisect_left(v, x)
exists = x in v
```

Prefix sums / Accumulation

```
vector<int> pre(n+1);
for(int i=0;i<n;i++) pre[i+1] = pre[i]+v[i];
int total = accumulate(v.begin(), v.end(), 0);
```

```
pre = [0]
for x in v: pre.append(pre[-1]+x)
total = sum(v)
```

Rotate

```
rotate(v.begin(), v.begin()+k, v.end());
```

```
v = v[k:] + v[:k]
```

Set / Multiset

```
set<int> s; // unique, ordered
s.insert(x); s.erase(x);
multiset<int> ms; // allows duplicates
ms.insert(x); ms.erase(ms.find(x));
```



```
s = set()
s.add(x); s.discard(x)
from collections import Counter
ms = Counter()
ms[x]+=1; ms[x]-=1;
if ms[x]==0: del ms[x]
```

Map / Dictionary

```
unordered_map<int,int> mp;
mp[x] = y;
if(mp.count(x)) ...
```

```
mp = {}
mp[x] = y
if x in mp:
    ...
```

Priority Queue / Heap

```
priority_queue<int> pq; // max-heap
priority_queue<int, vector<int>, greater<int>> pq; // min-heap
pq.push(x); pq.pop(); pq.top();
```

```
import heapq
pq = []
heapq.heappush(pq, -x) # max-heap
x = -heapq.heappop(pq)
heapq.heappush(pq, x) # min-heap
x = heapq.heappop(pq)
```

Queue / Deque

```
queue<int> q;
q.push(x); q.pop(); q.front(); q.back();
deque<int> dq;
dq.push_front(x); dq.pop_back();
```

```
from collections import deque
q = deque()
q.append(x); q.popleft()
dq = deque()
dq.appendleft(x); dq.pop()
```

Graph / Adjacency List

```
vector<vector<int>> g(n);
g[u].push_back(v); // directed

g = [[] for _ in range(n)]
g[u].append(v)
```

Bit tricks / Masks

```
(x>>k)&1 // check bit
x |= (1<<k) // set
x &= ~(1<<k) // clear
x ^= (1<<k) // toggle
__builtin_popcount(x) // number of 1s in binary
```

```
(x>>k)&1
x |= (1<<k)
x &= ~(1<<k)
x ^= (1<<k)
bin(x).count('1') # popcount
x.bit_count() # Python 3.10+
```

8.2 STL

```
#include <bits/stdc++.h>
using namespace std;
typedef vector<int> vi;
typedef vector<vi> vvi;
```

```
typedef pair<int,int> pii;
typedef vector<pii> vpi;
int main()
{
    // fast IO
    ios::sync_with_stdio(false);
    cin.tie(NULL);
    double pi = 3.14159; int x = 12; bool flag = false;
    // Precision / floating point
    cout << fixed << setprecision(2) << pi << '\n';
    // Width / alignment
    cout << setw(5) << x << '\n'; // pad to 5 chars
    cout << left << setw(5) << x; // left-aligned
    cout << right << setw(5) << x; // right-aligned
    // Fill character
    cout << setfill('0') << setw(5) << x << '\n'; // 00012
    // Bool as true/false
    cout << boolalpha << flag << '\n';
    cout << noboolalpha << flag << '\n'; // 0 or 1
    // Hex / Oct / Dec
    cout << hex << x << '\n';
    cout << oct << x << '\n';
    cout << dec << x << '\n';
    // Show sign
    cout << showpos << x << '\n';
    cout << noshowpos << x << '\n';
    // Lambda functions
    int a;
    auto f = [](){}; // empty capture: no outer variables, only parameters +
    ↪ globals
    auto f = [=](int x){cout <<a;}; // capture all outer variables by value
    auto f = [&](int x){a=2;}; // capture all outer variables by reference
    // Sort using custom function
    vpi P = {{1,2},{3,0},{2,5}};
    sort(P.begin(), P.end(), [](auto &a, auto &b){ return a.second < b.second; });
    vi vec = {1,2,3};
    sort(vec.begin(), vec.end()); // sort ascending
    sort(vec.rbegin(), vec.rend()); // descending
    // Fills v with sequentially increasing values, starting from 0
    vi v(10); iota(v.begin(), v.end(), 0);
    // string containing 10 copies of 'a'
    string a(10, 'a');
    // String-int conversion
    int i = stoi("42");
    string s = to_string(i);
    const char * cstr = s.c_str(); // string to c style string
    // transform string to lowercase
    string t = "HOLA";
    transform(t.begin(), t.end(), t.begin(), ::tolower);
    string spaces = " a a jdsa";
    // Remove leading spaces
    spaces.erase(spaces.begin(), find_if(spaces.begin(), spaces.end(),
    [](auto c){ return !isspace(c); }));

    // Remove all spaces
    spaces.erase(remove(spaces.begin(), spaces.end(), ' '), spaces.end());
    // Remove all tabs, newlines, spaces, etc...
    spaces.erase(remove_if(spaces.begin(), spaces.end(),
    [](auto c){ return isspace(c); }),
    spaces.end());
    // Split
    vector<string> tok; string tmp;
    stringstream ss(s);
    while (ss >> tmp) tok.push_back(tmp);
    // Pair priority queues using lexicographical order.
```

```
priority_queue<pii, vpi , greater<pii>> min_pq;
priority_queue<pii> max_pq;
// Min max element
auto [mn, mx] = minmax_element(v.begin(), v.end());
// Unique
```

```
sort(v.begin(), v.end());
v.erase(unique(v.begin(), v.end()), v.end());
int x = clamp(x, -1, 1);
}
```